

Dental OCT: A Primer on the Domain, Terminology, and AI State of the Art

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Optical coherence tomography (OCT) is well established in ophthalmology but still emerging in dentistry, where it can image teeth and oral soft tissue at micrometre resolution without radiation. This primer covers the dental anatomy and clinical vocabulary needed to work in the area, how OCT behaves on dental tissue, where AI in dental imaging stands today, and what little has been done with AI on dental OCT specifically — which is where the novel openings are.

1. Tooth and mouth anatomy

A tooth has two visible parts: the **crown** (above the gum) and the **root** (anchored in bone). From the outside in:

- **Enamel** — the outer layer of the crown; the hardest tissue in the body (~96% mineral, hydroxyapatite crystals arranged in *enamel rods/prisms*). No living cells, so it cannot regrow once cavitated, though early mineral loss can be reversed. Roughly 1–2.5 mm thick. Highly transparent near 1300 nm, which is why OCT works so well on it.
- **Dentin** — beneath the enamel; softer (~70% mineral), traversed by microscopic **dentinal tubules**. Scatters light more strongly than enamel.
- **Dentin-enamel junction (DEJ)** — the boundary between the two. A key OCT landmark: whether a lesion has crossed the DEJ is often what decides treatment.
- **Pulp** — the soft core containing nerves and blood vessels.
- **Cementum** — thin layer covering the root.

Supporting (periodontal) tissues:

- **Gingiva** — the gum.
- **Gingival sulcus** — the small groove between gum and tooth; when it deepens through disease it becomes a **periodontal pocket**.
- **Periodontal ligament** and **alveolar bone** — hold the tooth in its socket.
- **Oral mucosa** — the lining of cheeks, tongue, and floor of mouth; made of an **epithelium** on top of the **lamina propria**, separated by the **basement membrane**. Breach of the basement membrane is what distinguishes invasive cancer.

Tooth surfaces and types (used to describe where a lesion is):

- **Occlusal** — the chewing surface of back teeth, with **pits and fissures** where hidden caries often start.
- **Proximal / interproximal** — the surfaces between neighbouring teeth; hard to see directly.
- **Buccal / labial** (cheek/lip side), **lingual / palatal** (tongue/palate side), **cervical** (near the gum line).
- **Anterior** teeth (incisors, canines) vs **posterior** teeth (premolars, molars). Most intraoral OCT probes reach anterior teeth easily; posterior and proximal surfaces are harder.

2. The dental conditions that matter

Caries (tooth decay)

The most common disease worldwide. Bacteria in **plaque/biofilm** produce acids that dissolve mineral from enamel (**demineralization**). If the balance tips back — saliva, fluoride — mineral can return (**remineralization**).

- **White spot lesion** — the earliest visible stage: chalky enamel with subsurface mineral loss but an intact surface. Reversible.
- **Non-cavitated vs cavitated** — whether the surface has physically broken down. Non-cavitated lesions are usually managed non-invasively (fluoride, sealants); cavitated lesions usually need a filling.
- **Enamel caries vs dentin caries** — depth of the lesion; crossing the DEJ is a key threshold.
- **Active vs arrested** lesions — whether decay is progressing.
- **Secondary (recurrent) caries** — decay at the margin of an existing filling; a major reason restorations are replaced.
- **Root caries** — decay on exposed root surfaces, common in older adults.
- **ICDAS** (International Caries Detection and Assessment System) — the standard 0–6 visual scoring scale, from sound (0) to extensive cavity (6). Often used as a clinical label.
- **Minimally invasive dentistry** — the modern philosophy of detecting lesions early and arresting them rather than drilling. This is the clinical case for OCT: it sees early subsurface lesions that X-rays miss.

Restorations

- **Restoration** — any filling or crown. **Composite resin** (tooth-coloured) is the most common filling today.
- **Marginal integrity / marginal gap** — how well the filling seals against the tooth. **Microleakage** and **debonding** (gaps at the tooth–resin interface) let bacteria in and cause secondary caries. OCT can see these interfaces directly.

Periodontal disease

- **Gingivitis** — reversible gum inflammation.
- **Periodontitis** — destructive disease with loss of attachment and bone; staged and graded clinically. Measured by **probing depth** and **clinical attachment loss**.
- **Calculus (tartar)** — hardened plaque, including below the gum (**subgingival calculus**).

Oral cancer and precancer

- **Oral squamous cell carcinoma (OSCC)** — the dominant oral cancer.
- **Oral potentially malignant disorders (OPMD)** — lesions like **leukoplakia** (white patch) and **erythroplakia** (red patch) that can progress.
- **Dysplasia** — abnormal cell growth graded mild/moderate/severe; diagnosed by **biopsy** and **histopathology**, the ground truth.

Other

- **Enamel cracks / cracked-tooth syndrome, erosion** (acid wear, not bacterial), **abrasion**, and **fluorosis**.

3. Dental imaging modalities and where OCT fits

Modality	What it is	Strengths	Limits
Bitewing radiograph	Intraoral X-ray of crowns of upper and lower back teeth	Standard for proximal caries	Radiation; misses early enamel lesions; 2D
Periapical radiograph	Intraoral X-ray of whole tooth and root tip	Root and bone problems	Radiation; 2D

Modality	What it is	Strengths	Limits
Panoramic radiograph (OPG)	Single wide X-ray of both jaws	Overview screening	Low detail; distortion
CBCT	Cone-beam CT, 3D X-ray	3D bone, implants, roots	Higher radiation; ~80–400 μm resolution; poor for soft tissue and microcracks
Intraoral photographs / scanners	Colour images and 3D surface scans	Cheap, ubiquitous	Surface only
NILT / NIRI	Near-infrared transillumination (e.g. DIAGNOcam, iTero NIRI)	Radiation-free caries screening	2D, no depth measurement
Fluorescence (QLF, laser fluorescence)	Detects mineral loss or bacterial fluorescence	Early lesion signal	Indirect, no structure
Micro-CT	Lab-only high-resolution X-ray CT	Gold-standard mineral ground truth for extracted teeth	Not usable in patients
Histology	Sectioned and stained tissue	Ground truth for cancer and lesion depth	Destructive
OCT	Near-infrared interferometric cross-sectional imaging	~10–15 μm resolution, depth 0.5–2 mm, no radiation, real-time, 3D	Shallow depth; hardware not yet in routine clinics; limited access to posterior teeth

OCT's niche: **depth-resolved, radiation-free, micron-scale imaging of the first ~2 mm** — exactly the zone where early caries, restoration margins, cracks, gingiva, and mucosal epithelium live.

4. How OCT behaves on dental tissue

- **Wavelength.** Dental OCT mostly uses ~1300 nm, where enamel is highly transparent, allowing imaging through the full enamel thickness to the DEJ. (Retinal OCT uses ~840 or ~1050 nm.)
- **Systems.** Time-domain, spectral-domain, and swept-source OCT have all been used; **swept-source** dominates modern dental prototypes for speed and depth.
- **Polarization-sensitive OCT (PS-OCT) and cross-polarization OCT (CP-OCT).** Demineralized enamel scrambles (depolarizes) polarized light, so lesions appear bright in the cross-polarized channel while the strong surface reflection is suppressed. This gives much better lesion contrast and allows quantifying severity by **integrated reflectivity** over lesion depth.
- **What appears in a dental B-scan.** Sound enamel is dark/low-scatter; demineralized enamel scatters strongly and appears bright; the DEJ shows as a boundary line; dentin is more uniformly scattering; gaps under restorations show as bright lines; mucosal epithelium and lamina propria appear as distinct layers whose boundary disappears in invasive cancer.
- **Key quantitative readouts.** Lesion depth, integrated reflectivity (ΔR), enamel thickness, remaining enamel above the DEJ, gap length at restoration margins, epithelial thickness, and attenuation coefficient.
- **Ground truth.** Micro-CT or transverse microradiography (TMR) for mineral loss on extracted teeth; histology for lesion depth and cancer; ICDAS or expert consensus in vivo.
- **Ex vivo vs in vivo.** Most studies use extracted teeth (ex vivo). In the mouth, saliva, motion, soft tissue, and probe access add a real domain shift.

5. Clinical applications of dental OCT

- **Early caries detection and monitoring** — the flagship use. OCT detects subsurface demineralization before cavitation, including hidden occlusal caries, and can monitor lesions over time and response to fluoride.
- **Restoration assessment** — marginal gaps, debonding, secondary caries under composite fillings.
- **Periodontal imaging** — gingival thickness, sulcus, calculus, plaque.
- **Oral cancer and precancer screening** — epithelial thickening and basement-membrane disruption as non-invasive biopsy guidance.

- **Cracks, erosion, enamel thickness**, and emerging intraoral-scanner integration (crown margin detection, 3D tooth geometry through blood and saliva).
- **Commercial movement** — handheld intraoral OCT scanners with AI analysis (e.g. Perceptive) and combined intraoral scan + OCT devices (e.g. Lillivis OCTiX) are reaching the market, which signals that clinical-scale dental OCT data will grow.

6. AI in dental imaging: the broader state of the art

AI in dentistry is dominated by **X-ray and photograph** data, not OCT.

- **Caries on radiographs** — the most-studied task. Reviews report AUCs ranging roughly 0.74–0.99 for caries diagnosis, with segmentation performance around 0.84–0.86. A recent end-to-end system on 1,200 radiovisiography images used U-Net for lesion segmentation (Dice 0.89) and ResNet-50 for three-class grading (no caries / enamel / dentin) at 93.2% accuracy.
- **Caries on intraoral photos** — CNN classifiers of enamel caries stages on public photo datasets (~2,000 images) with explainability.
- **Near-infrared transillumination** — ResNet caries models trained in vitro and in vivo, with explicit tests of how poorly models generalize between the two settings.
- **Other common tasks** — tooth numbering and segmentation on panoramic X-rays and CBCT, periodontal bone-loss staging, periapical lesion detection, cephalometric landmark detection, and oral-cancer screening from photographs.
- **Commercial AI** — FDA/CE-cleared radiograph-analysis tools already tag caries, bone loss, and restorations for dentists; they are decision support, not autonomous.

Across this literature the recurring weaknesses are small single-centre datasets, inconsistent labels, limited external validation, and poor generalization across devices and settings.

7. AI on dental OCT specifically

This is a thin literature — the gap that makes it attractive.

- **Caries classification** — CNNs (e.g. ResNet-152) on ex vivo dental OCT have reported around 95% accuracy for caries detection, outperforming clinicians in that setting. Another study built a three-grade system (healthy / enamel caries / dentin caries) on extracted teeth imaged with both spectral-domain and swept-source OCT, comparing ten CNN designs.

- **Lesion detection** — transformer-based detectors have been tested on dental OCT.
- **Oral cancer** — CNNs on OCT of normal mucosa, precancer, and OSCC have reached ~96.8% accuracy in-house.
- **Industry** — intraoral OCT scanners with built-in AI report ~90% caries-detection accuracy.
- **Character of the evidence** — almost all studies are small (tens to low hundreds of teeth), ex vivo, single-device, and use private data. **No public dental-OCT dataset or benchmark exists.** Classical image processing (thresholding, attenuation analysis, integrated reflectivity) is still common.

8. Where novel work is possible

- **The first public, annotated dental-OCT dataset and benchmark** — on its own a meaningful contribution.
- **Caries depth staging with a treatment-decision output** (monitor/remineralize vs restore) — a genuine clinical decision support target.
- **Segmentation of dental structures** — enamel, DEJ, dentin, lesion extent, restoration gaps, epithelium — enabling quantitative biomarkers rather than just labels.
- **3D volumetric analysis** rather than single B-scans.
- **Self-supervised pretraining and transfer from retinal OCT** — same physics, abundant data.
- **Uncertainty and out-of-distribution detection** — flagging uninformative or unfamiliar scans, essential for a CDSS.
- **Ex vivo → in vivo and cross-device domain adaptation.**
- **Generative denoising and data synthesis** with checks that lesion signal is preserved.
- **Multimodal fusion** of OCT with radiographs, NIR, or photos.
- **Oral-cancer triage** from mucosal OCT, validated on histology.

9. Quick glossary

Term	Meaning
Caries	Tooth decay caused by bacterial acid dissolving mineral

Term	Meaning
Demineralization / remineralization	Mineral loss / mineral regain in enamel or dentin
White spot lesion	Earliest visible, reversible enamel caries
Cavitation	Physical breakdown of the tooth surface
DEJ	Dentin–enamel junction
ICDAS	0–6 visual caries scoring system
Occlusal / proximal / buccal / lingual	Chewing / between-teeth / cheek-side / tongue-side surfaces
Pits and fissures	Grooves on chewing surfaces where decay hides
Composite resin	Tooth-coloured filling material
Secondary caries	Decay at a filling's margin
Microleakage / debonding	Gaps between filling and tooth
Gingivitis / periodontitis	Reversible gum inflammation / destructive gum and bone disease
Calculus	Hardened plaque (tartar)
OSCC	Oral squamous cell carcinoma
OPMD	Oral potentially malignant disorders (e.g. leukoplakia)
Dysplasia	Abnormal, potentially precancerous cell growth
Bitewing / periapical / panoramic	Types of dental X-ray
CBCT	Cone-beam CT, 3D dental X-ray
NILT / NIRI	Near-infrared transillumination imaging
PS-OCT / CP-OCT	Polarization-sensitive / cross-polarization OCT
Integrated reflectivity (ΔR)	OCT measure of lesion severity
Micro-CT / TMR	Lab gold standards for mineral content
Ex vivo / in vivo	On extracted teeth / in the living patient

Term	Meaning
Minimally invasive dentistry	Detecting early and arresting decay rather than drilling

Sources

- [Daniel Fried, UCSF — dental OCT publications](#)
- [Clinical assessment of early tooth demineralization using PS-OCT](#)
- [Imaging caries lesions and progression with PS-OCT](#)
- [NIH: light-based technologies for detecting dental decay](#)
- [Diagnosis of occlusal caries with 3D swept-source OCT \(Yoshida prototype\)](#)
- [SS-OCT for 3D abutment tooth data \(dental SS-OCT applications review\)](#)
- [Portable TD-OCT scanner for dental use \(2026\)](#)
- [OCTiX intraoral OCT scanner \(Lilivis/Huvitz\)](#)
- [Dental technologies for 2025, including intraoral OCT \(Inside Dentistry\)](#)
- [Three-grade deep learning caries classification on SD- and SS-OCT](#)
- [End-to-end deep learning for caries on radiovisiography images](#)
- [AI in dental caries diagnosis: umbrella review](#)
- [Generalizability of deep learning caries models on NILT images](#)
- [Automated enamel caries classification from intraoral images](#)